SMART INDIA HACKATHON 2026

Problem Statement ID: SIH26251

Problem Statement Title: Indian Army - Predictive Logistics & Forward Supply Chain

Theme: Transportation & Logistics

PS Category: Software

Team ID: 185972

Team Name: Subterra Mesh

Solution: DRONA-LOG: AI-driven predictive logistics and terrain-aware supply optimisation



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PROPOSED SOLUTION

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The problem

Demand, stock, transport capacity and terrain/weather data sit in separate streams, so static plans fail when a pass closes.

Our solution

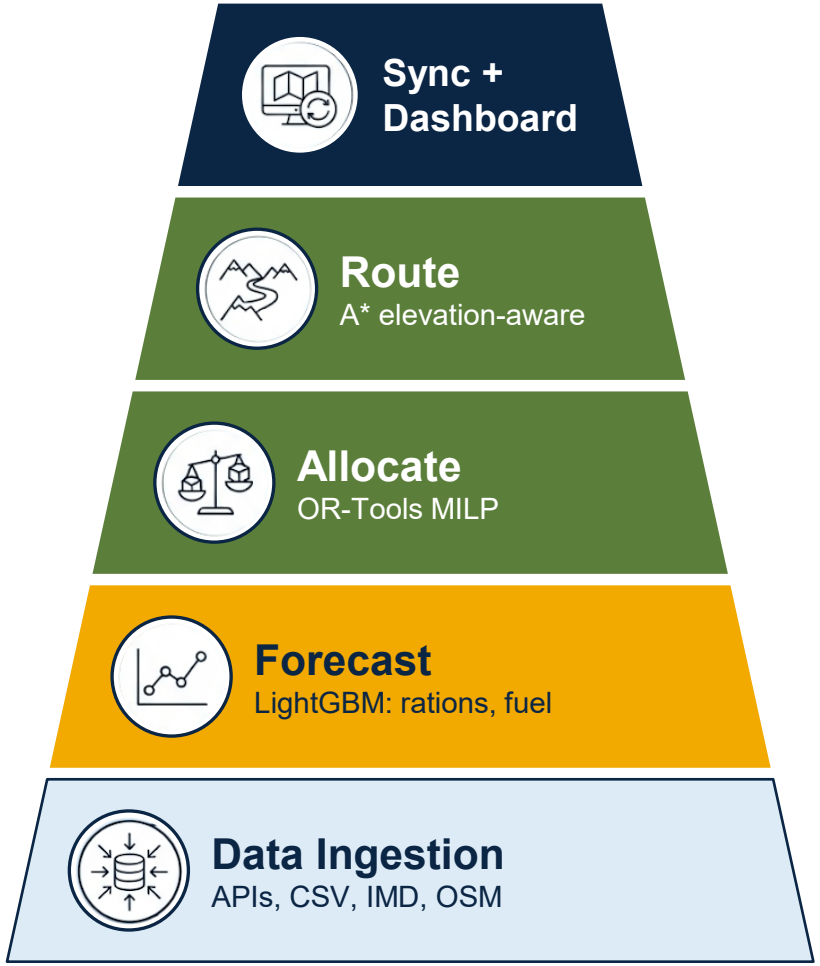
DRONA-LOG forecasts ration and fuel demand, optimises allocation and re-routes supplies in one integrated system.

What is different

Map-first UI, terrain-aware routing, closed-loop re-planning, and an offline-first field client that syncs when the network returns.

▶ Demo Video
2:33 narrated walkthrough

Prototype
GitHub · run locally



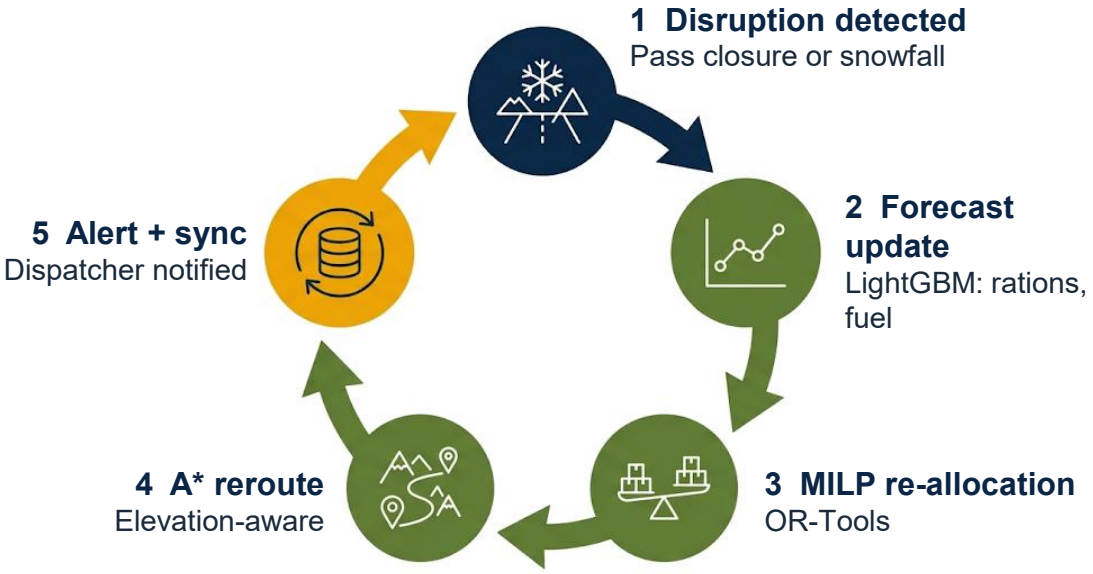
- Snowstorm / pass closure
- Dynamic rerouting
- Network outage
- Offline queue and sync
- Static schedules
- Predictive forecasting

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TECHNICAL APPROACH

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Methodology: closed-loop re-planning

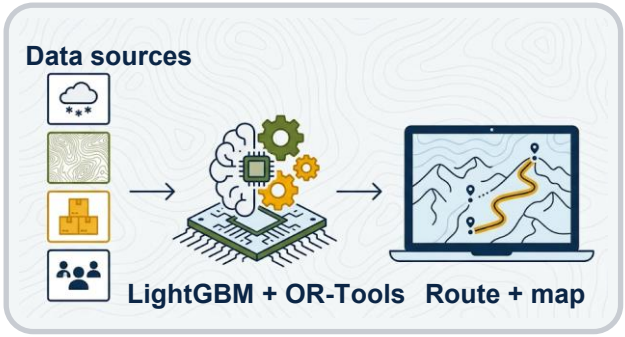


Measured: 15.4 ms reroute and 439 ms re-solve after Zojila closure (sample run).

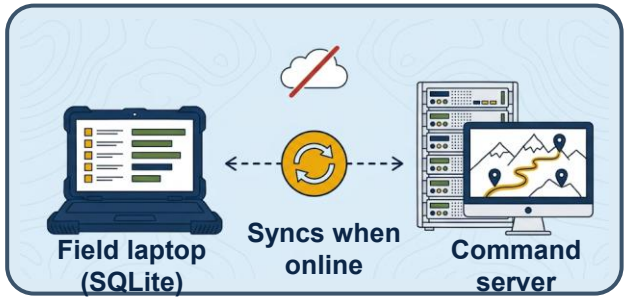
GitHub
Code · data · tests

Detailed Report
Architecture · API · tests

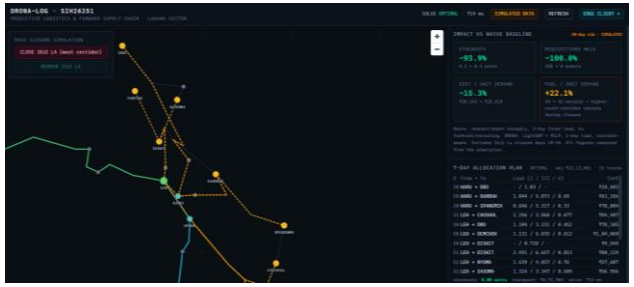
Data-to-decision pipeline



Offline-first sync



Field laptop works offline and syncs to the command server when the network returns.



Live command view — map, KPIs and plan

TECHNOLOGIES USED

Frontend
React · MapLibre GL · Vite

Backend (Python) and data
FastAPI · SQLite

AI and optimisation
LightGBM · OR-Tools · NetworkX

Deployment
Docker

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FEASIBILITY AND VIABILITY

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Technical Feasibility

- Runs on standard field laptops and private servers
- Open-source stack: LightGBM, OR-Tools, NetworkX, SQLite
- Optimisation enforces inventory, capacity and demand constraints
- Offline-first: local SQLite store syncs when the network returns

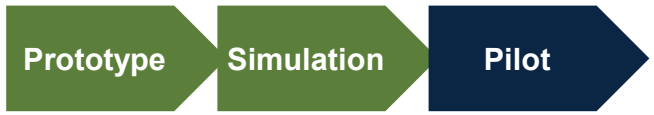
Operational Viability

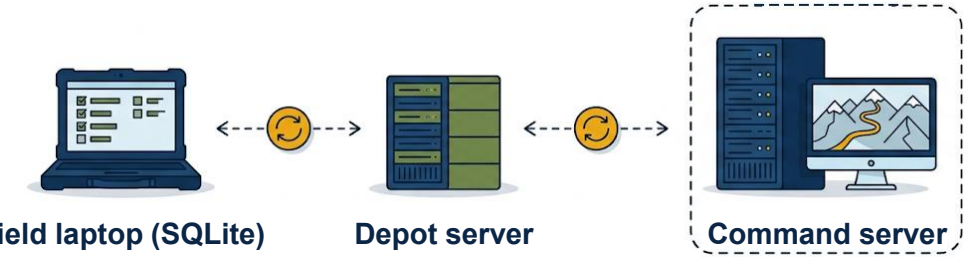
- Predicts shortages and triggers stock alerts early
- Traceable records and reliable synchronisation
- Fewer avoidable trips cut transport cost and fuel waste
- Extends to other remote, disruption-prone supply networks

Risks and Mitigation

Risk	Mitigation
No real military data	Synthetic data; pilot validation
Weak or no connectivity	Offline queue and sync
Data sensitivity	On-prem or air-gapped; encryption; role-based access
Rare-event model error	Safety-stock buffers; human override

Phased rollout





Security: the command server stays on-prem or air-gapped. Field laptops keep data locally and sync only when a link is permitted.

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IMPACT AND BENEFITS

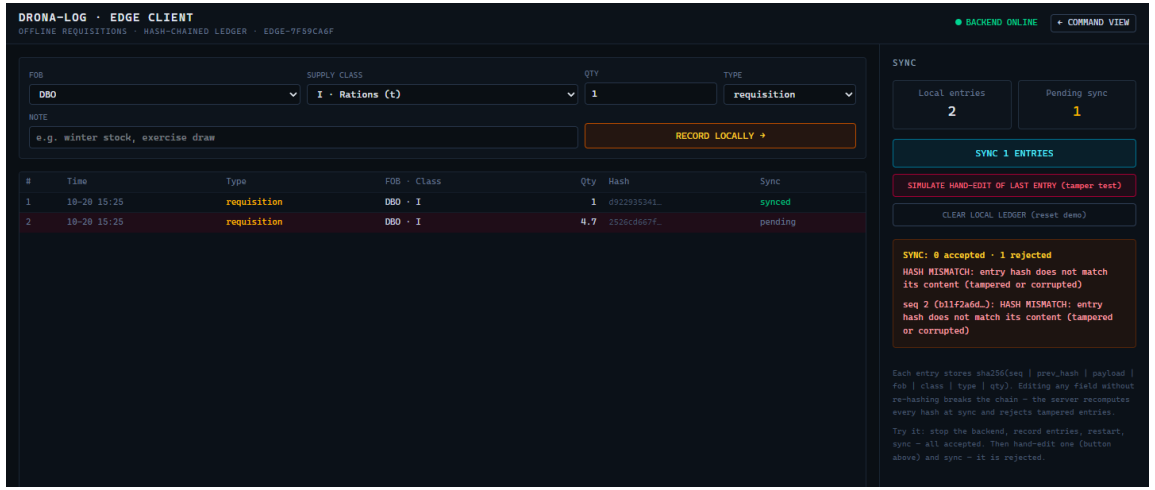
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Stockout prevention
Forecast demand and maintain safety stocks.

Faster decisions
Recalculate allocations and routes after disruptions.

Fuel efficiency
Optimise vehicle assignments and supply routes.

Offline continuity
Keep local records during network outages.



Live edge UI — offline ledger rejects a tampered entry

Who benefits: logistics officers, depot commanders, formation HQ and troops at forward posts.

439 ms — MILP re-solve after Zojila closure (sample run)

–93.9% — Stockout units vs naive baseline (30-day SIMULATED)

0 — Tampered ledger entries accepted by the server (SHA-256 verification)

Measured in the seeded prototype; operational figures SIMULATED.

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RESEARCH AND REFERENCES

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OR-Tools
Constrained inventory and routing models

LightGBM
Demand forecasting

NetworkX
Graph-based pathfinding

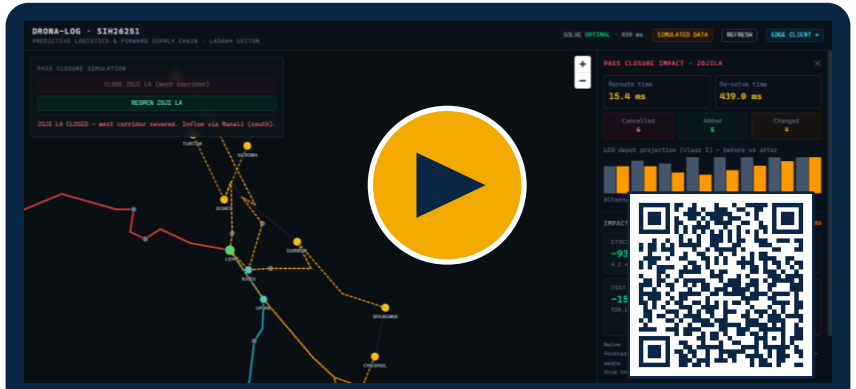
OSM / MapLibre
Geospatial data and maps

IMD
Weather inputs

FastAPI
Backend service

Research papers

1. **Ke et al.** LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS 2017. [PDF]
2. **Reich et al.** A framework for optimizing sustainment logistics for a US Army infantry brigade combat team. J. Defense Analytics and Logistics 4(2), 2020. [DOI]
3. **Liu et al.** Improved A* Algorithm for Mobile Robots under Rough Terrain Based on Ground Trafficability Model and Ground Ruggedness Model. Sensors 24(15), 2024. [DOI]



▸ **Demo video** *(2:33 narrated walkthrough)*

The 2:33 narrated demo shows a Zojila closure, Manali reroute, offline requisition sync and a rejected hash-chain tamper.

GitHub
Code · data · tests

Detailed Report
Architecture · API · tests

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